Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering
Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory
Academic Year	2025–2026 (Even)
Batch	2023–2027
Name	Sabarithan P

Experiment 1: Working with Python packages – NumPy, SciPy, Scikit-learn, Matplotlib

Aim:
To study and explore Python library packages such as Pandas, NumPy, Matplotlib, Scikit-learn, and SciPy to understand data science and machine learning concepts, and to analyze how datasets are explored and mapped to machine learning models using exploratory data analysis techniques.

Libraries Used:

- **NumPy:** Numerical computations and multi-dimensional array handling.
- **Pandas:** Data manipulation, cleaning, and analysis using DataFrames.
- **Matplotlib:** Visualization using line plots, bar charts, and histograms.
- **Seaborn:** Advanced statistical data visualization.
- **Scikit-learn:** Machine learning algorithms, preprocessing, training, and evaluation.
- **SciPy:** Scientific computing, optimization, and statistical analysis.

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January 26, 2026

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
↪cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, ConfusionMatrixDisplay,
    silhouette_score, davies_bouldin_score, adjusted_rand_score,
    classification_report
)
from sklearn.decomposition import PCA
from sklearn.feature_selection import SelectKBest, f_classif
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
from sklearn.cluster import KMeans

RANDOM_STATE = 42
```

#IRIS DATASET

[2]:
```
df = pd.read_csv("Iris.csv")
df.head()
```

[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

[3]:
```python
print("Shape:", df.shape)

print("\nInfo:")
df.info()

print("\nSpecies counts:")
print(df["Species"].value_counts())
```

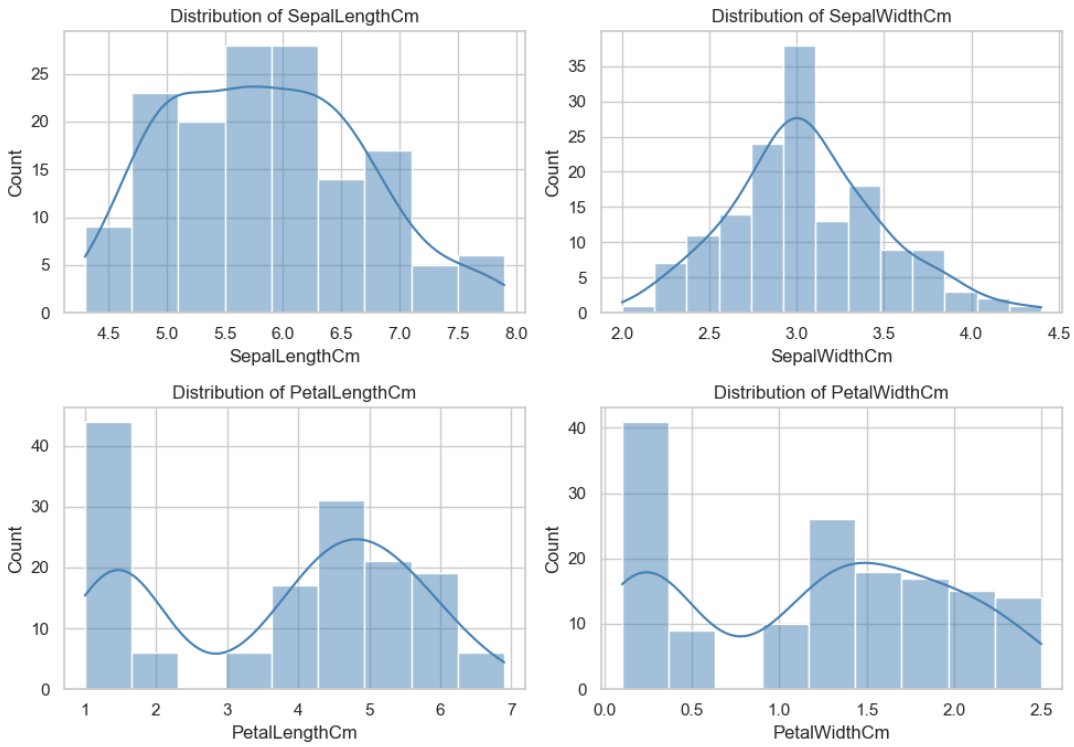
```
Shape: (150, 6)

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB

Species counts:
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

[4]:
```python
features = ["SepalLengthCm", "SepalWidthCm", "PetalLengthCm", "PetalWidthCm"]
df[features].describe()
```

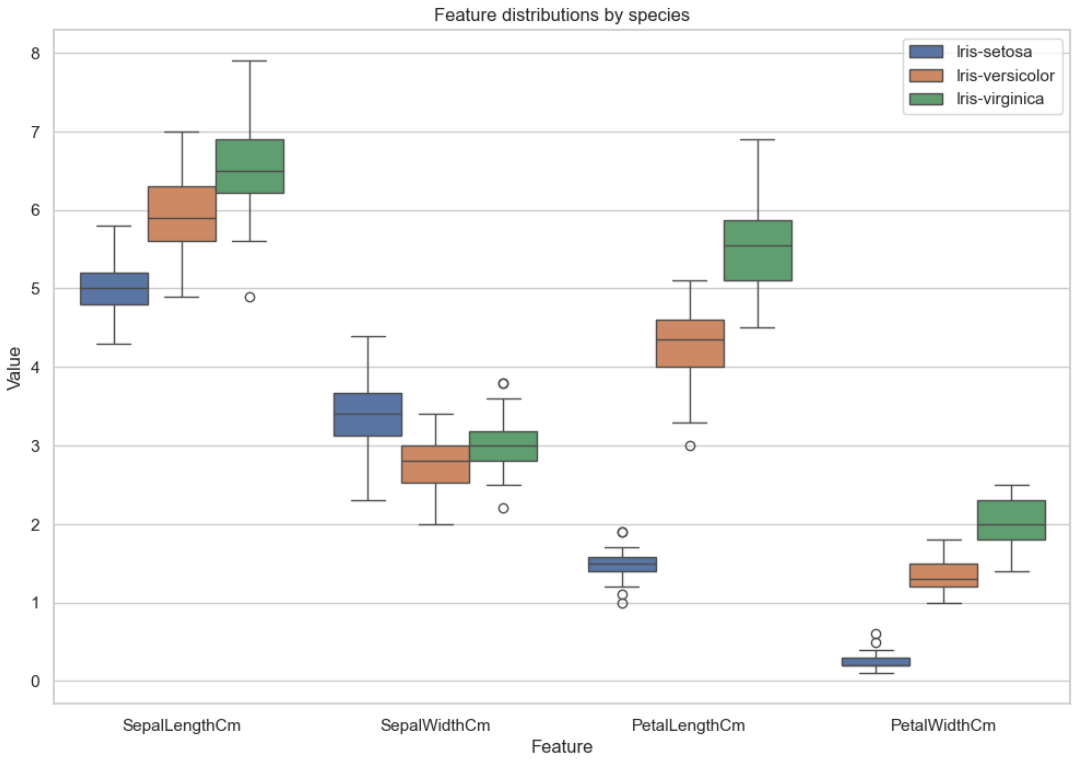
[4]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

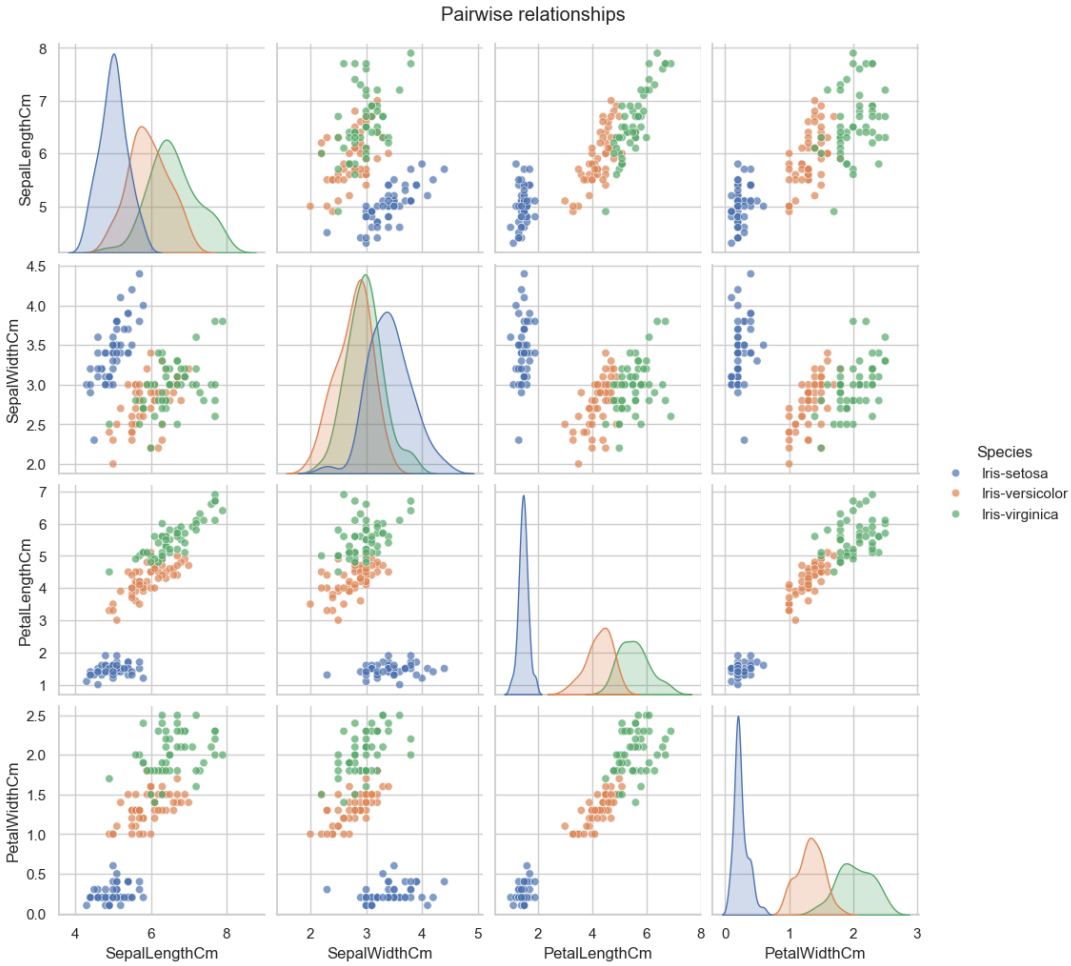
[5]:
```
sns.set(style="whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(10, 7))
for ax, col in zip(axes.ravel(), features):
    sns.histplot(df[col], kde=True, ax=ax, color="steelblue")
    ax.set_title(f"Distribution of {col}")

plt.tight_layout()
plt.show()
```



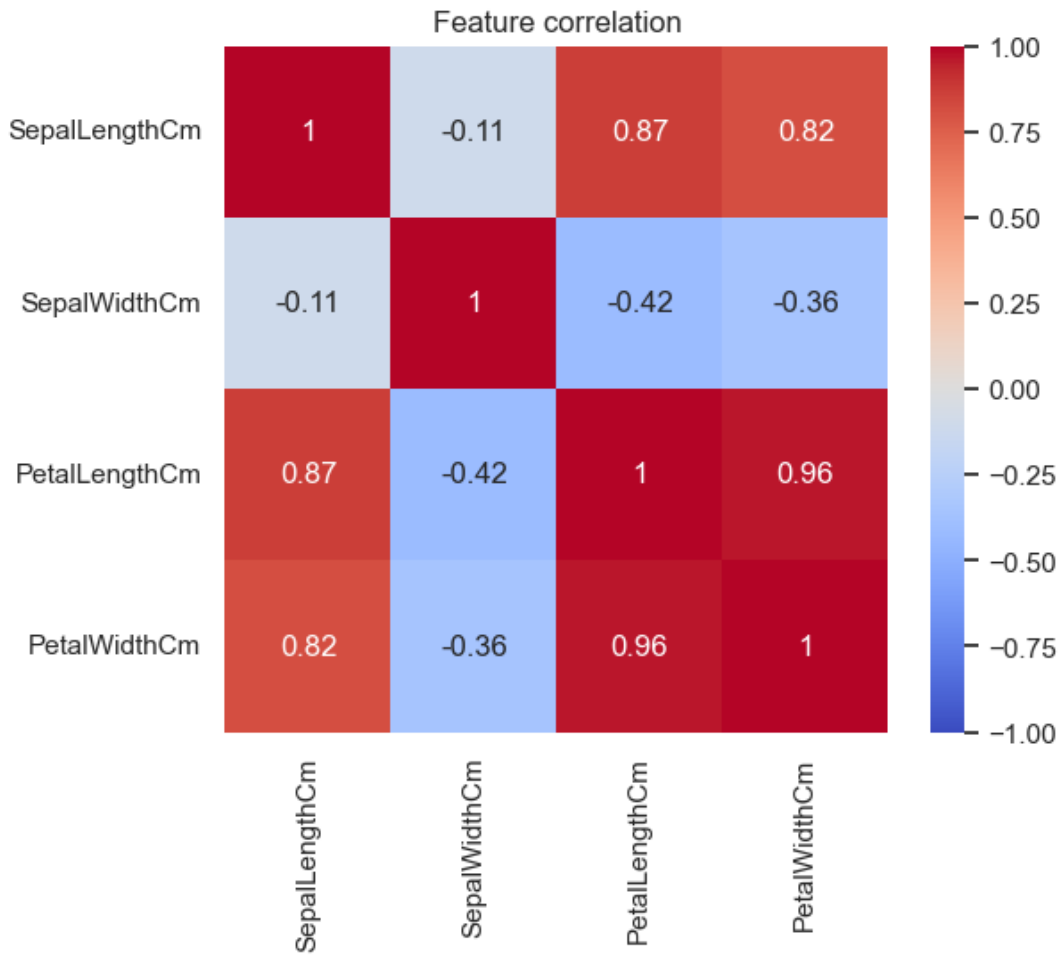
[6]:
```
plt.figure(figsize=(12, 8))

melted = df.melt(
    id_vars="Species",
    value_vars=features,
    var_name="Feature",
    value_name="Value"
)

sns.boxplot(data=melted, x="Feature", y="Value", hue="Species")
plt.title("Feature distributions by species")
plt.legend(loc="best")
plt.show()
```



[7]:
```
sns.pairplot(
    df,
    hue="Species",
    vars=features,
    diag_kind="kde",
    plot_kws={"alpha": 0.7}
)

plt.suptitle("Pairwise relationships", y=1.02)
plt.show()
```



[8]:
```
plt.figure(figsize=(6, 5))

corr = df[features].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", vmin=-1, vmax=1)

plt.title("Feature correlation")
plt.show()
```



#LOAN APPROVAL DATASET

[2]:
```
df = pd.read_csv("loan_train.csv")
df.head()
```

[2]:
```
  Customer ID               Name Gender  Age  Income (USD) Income Stability  \
0     C-36995   Frederica Shealy      F   56       1933.05              Low
1     C-33999  America Calderone      M   32       4952.91              Low
2      C-3770      Rosetta Verne      F   65        988.19             High
3     C-26480         Zoe Chitty      F   65           NaN             High
4     C-23459       Afton Venema      F   31       2614.77              Low

  Profession     Type of Employment    Location  Loan Amount Request (USD)  \
0    Working            Sales staff  Semi-Urban                   72809.58
1    Working                    NaN  Semi-Urban                   46837.47
2  Pensioner                    NaN  Semi-Urban                   45593.04
3  Pensioner                    NaN       Rural                   80057.92
4    Working  High skill tech staff  Semi-Urban                  113858.89
```

```
   ...  Credit Score No. of Defaults Has Active Credit Card  Property ID  \
0  ...        809.44               0                    NaN          746
1  ...        780.40               0            Unpossessed          608
2  ...        833.15               0            Unpossessed          546
3  ...        832.70               1            Unpossessed          890
4  ...        745.55               1                 Active          715

   Property Age  Property Type Property Location  Co-Applicant  \
0       1933.05              4             Rural             1
1       4952.91              2             Rural             1
2        988.19              2             Urban             0
3           NaN              2        Semi-Urban             1
4       2614.77              4        Semi-Urban             1

   Property Price  Loan Sanction Amount (USD)
0       119933.46                    54607.18
1        54791.00                    37469.98
2        72440.58                    36474.43
3       121441.51                    56040.54
4       208567.91                    74008.28

[5 rows x 24 columns]
```

[3]:
```python
print("Shape:", df.shape)

print("\nInfo:")
df.info()

print("\nMissing values per column:")
print(df.isnull().sum())
```

```
Shape: (30000, 24)

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30000 entries, 0 to 29999
Data columns (total 24 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Customer ID                 30000 non-null  object
 1   Name                        30000 non-null  object
 2   Gender                      29947 non-null  object
 3   Age                         30000 non-null  int64
 4   Income (USD)                25424 non-null  float64
```

```
 5   Income Stability             28317 non-null  object
 6   Profession                   30000 non-null  object
 7   Type of Employment           22730 non-null  object
 8   Location                     30000 non-null  object
 9   Loan Amount Request (USD)    30000 non-null  float64
 10  Current Loan Expenses (USD)  29828 non-null  float64
 11  Expense Type 1               30000 non-null  object
 12  Expense Type 2               30000 non-null  object
 13  Dependents                   27507 non-null  float64
 14  Credit Score                 28297 non-null  float64
 15  No. of Defaults              30000 non-null  int64
 16  Has Active Credit Card       28434 non-null  object
 17  Property ID                  30000 non-null  int64
 18  Property Age                 25150 non-null  float64
 19  Property Type                30000 non-null  int64
 20  Property Location            29644 non-null  object
 21  Co-Applicant                 30000 non-null  int64
 22  Property Price               30000 non-null  float64
 23  Loan Sanction Amount (USD)   29660 non-null  float64
dtypes: float64(8), int64(5), object(11)
memory usage: 5.5+ MB

Missing values per column:
Customer ID                       0
Name                              0
Gender                           53
Age                               0
Income (USD)                   4576
Income Stability               1683
Profession                        0
Type of Employment             7270
Location                          0
Loan Amount Request (USD)         0
Current Loan Expenses (USD)     172
Expense Type 1                    0
Expense Type 2                    0
Dependents                     2493
Credit Score                   1703
No. of Defaults                   0
Has Active Credit Card         1566
Property ID                       0
Property Age                   4850
Property Type                     0
Property Location               356
Co-Applicant                      0
```

```
Property Price                    0
Loan Sanction Amount (USD)      340
dtype: int64
```

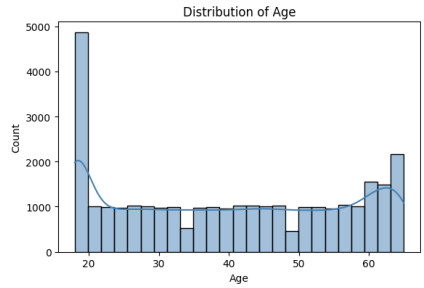
[4]:
```
df.describe(include="all")
```

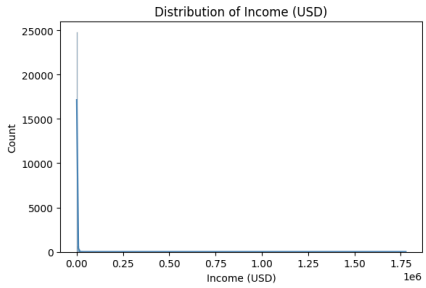
[4]:
```
       Customer ID              Name Gender           Age  Income (USD)  \
count        30000             30000  29947  30000.000000  2.542400e+04
unique       30000             30000      2           NaN           NaN
top        C-36995  Frederica Shealy      M           NaN           NaN
freq             1                 1  15053           NaN           NaN
mean           NaN               NaN    NaN     40.092300  2.630574e+03
std            NaN               NaN    NaN     16.045129  1.126272e+04
min            NaN               NaN    NaN     18.000000  3.777000e+02
25%            NaN               NaN    NaN     25.000000  1.650457e+03
50%            NaN               NaN    NaN     40.000000  2.222435e+03
75%            NaN               NaN    NaN     55.000000  3.090593e+03
max            NaN               NaN    NaN     65.000000  1.777460e+06

       Income Stability Profession Type of Employment    Location  \
count             28317      30000              22730       30000
unique                2          8                 18           3
top                 Low    Working           Laborers  Semi-Urban
freq              25751      16926               5578       21563
mean                NaN        NaN                NaN         NaN
std                 NaN        NaN                NaN         NaN
min                 NaN        NaN                NaN         NaN
25%                 NaN        NaN                NaN         NaN
50%                 NaN        NaN                NaN         NaN
75%                 NaN        NaN                NaN         NaN
max                 NaN        NaN                NaN         NaN

        Loan Amount Request (USD)  ...  Credit Score No. of Defaults  \
count                30000.000000  ...  28297.000000    30000.000000
unique                        NaN  ...           NaN             NaN
top                           NaN  ...           NaN             NaN
freq                          NaN  ...           NaN             NaN
mean                 88826.333855  ...    739.885381        0.193933
std                  59536.949605  ...     72.163846        0.395384
min                   6048.240000  ...    580.000000        0.000000
25%                  41177.755000  ...    681.880000        0.000000
50%                  75128.075000  ...    739.820000        0.000000
75%                 119964.605000  ...    799.120000        0.000000
max                 621497.820000  ...    896.260000        1.000000
```

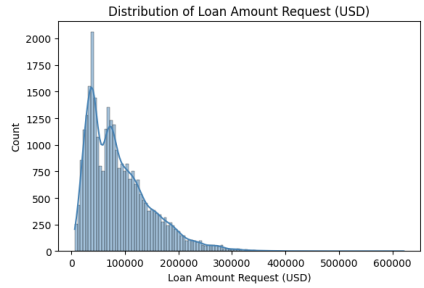
```
       Has Active Credit Card   Property ID  Property Age  Property Type  \
count                   28434  30000.000000  2.515000e+04   30000.000000
unique                      3           NaN           NaN            NaN
top                    Active           NaN           NaN            NaN
freq                     9771           NaN           NaN            NaN
mean                      NaN    501.934700  2.631119e+03       2.460067
std                       NaN    288.158086  1.132268e+04       1.118562
min                       NaN      1.000000  3.777000e+02       1.000000
25%                       NaN    251.000000  1.650450e+03       1.000000
50%                       NaN    504.000000  2.223250e+03       2.000000
75%                       NaN    751.000000  3.091408e+03       3.000000
max                       NaN    999.000000  1.777460e+06       4.000000

       Property Location  Co-Applicant  Property Price  \
count              29644  30000.000000    3.000000e+04
unique                 3           NaN             NaN
top           Semi-Urban           NaN             NaN
freq               10387           NaN             NaN
mean                 NaN     -4.743867    1.317597e+05
std                  NaN     74.614593    9.354955e+04
min                  NaN   -999.000000   -9.990000e+02
25%                  NaN      1.000000    6.057216e+04
50%                  NaN      1.000000    1.099936e+05
75%                  NaN      1.000000    1.788807e+05
max                  NaN      1.000000    1.077967e+06

        Loan Sanction Amount (USD)
count                 29660.000000
unique                         NaN
top                            NaN
freq                           NaN
mean                  47649.342208
std                   48221.146686
min                    -999.000000
25%                       0.000000
50%                   35209.395000
75%                   74261.250000
max                  481907.320000

[11 rows x 24 columns]
```

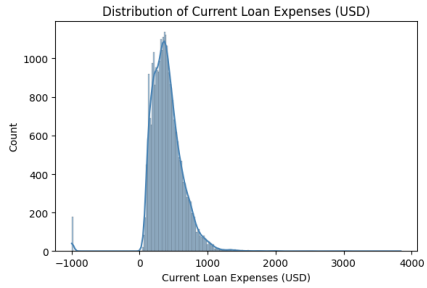
[5]:
```
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns

n_cols = 2
```

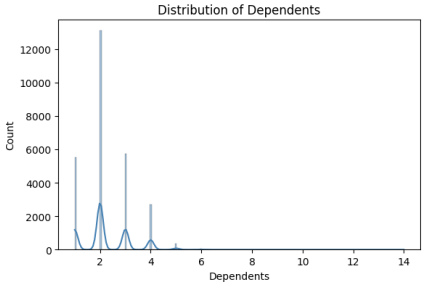
```python
n_rows = int(np.ceil(len(numeric_cols) / n_cols))

fig, axes = plt.subplots(n_rows, n_cols, figsize=(12, 4 * n_rows))
axes = axes.flatten()

for i, col in enumerate(numeric_cols):
    sns.histplot(df[col], kde=True, ax=axes[i], color="steelblue")
    axes[i].set_title(f"Distribution of {col}")

# Remove empty subplots (if any)
for j in range(i + 1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```

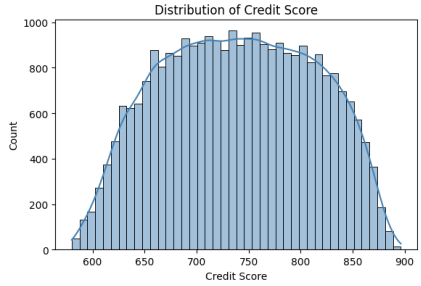

Distribution of Age
Count
Age

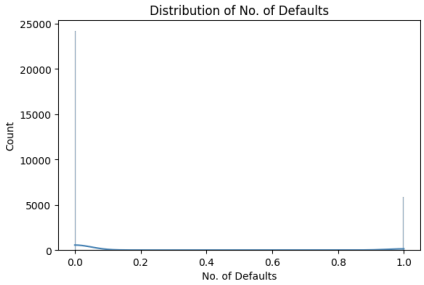

Distribution of Income (USD)
Count
Income (USD)
1e6

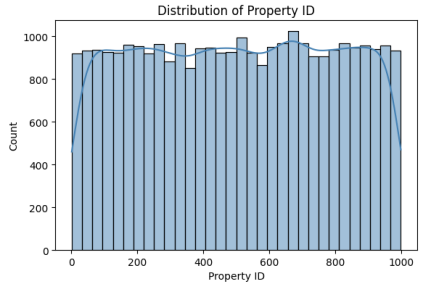

Distribution of Loan Amount Request (USD)
Count
Loan Amount Request (USD)

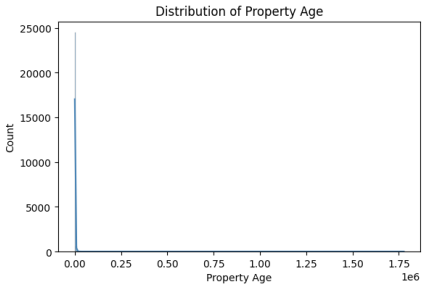

Distribution of Current Loan Expenses (USD)
Count
Current Loan Expenses (USD)

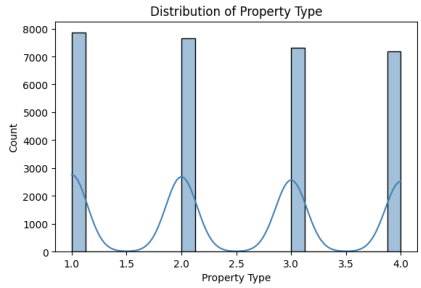

Distribution of Dependents
Count
Dependents

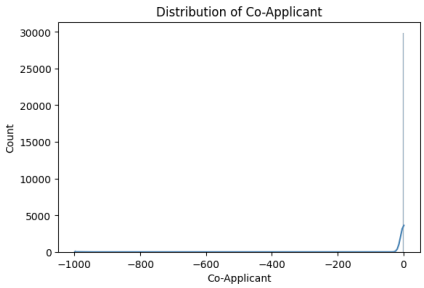

Distribution of Credit Score
Count
Credit Score


Distribution of No. of Defaults
Count
No. of Defaults

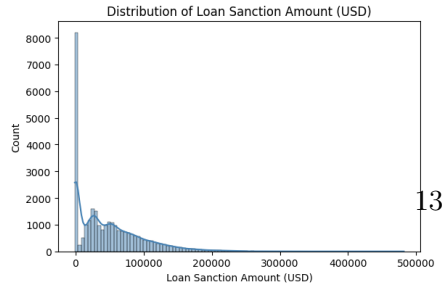

Distribution of Property ID
Count
Property ID


Distribution of Property Age
Count
Property Age
1e6


Distribution of Property Type
Count
Property Type

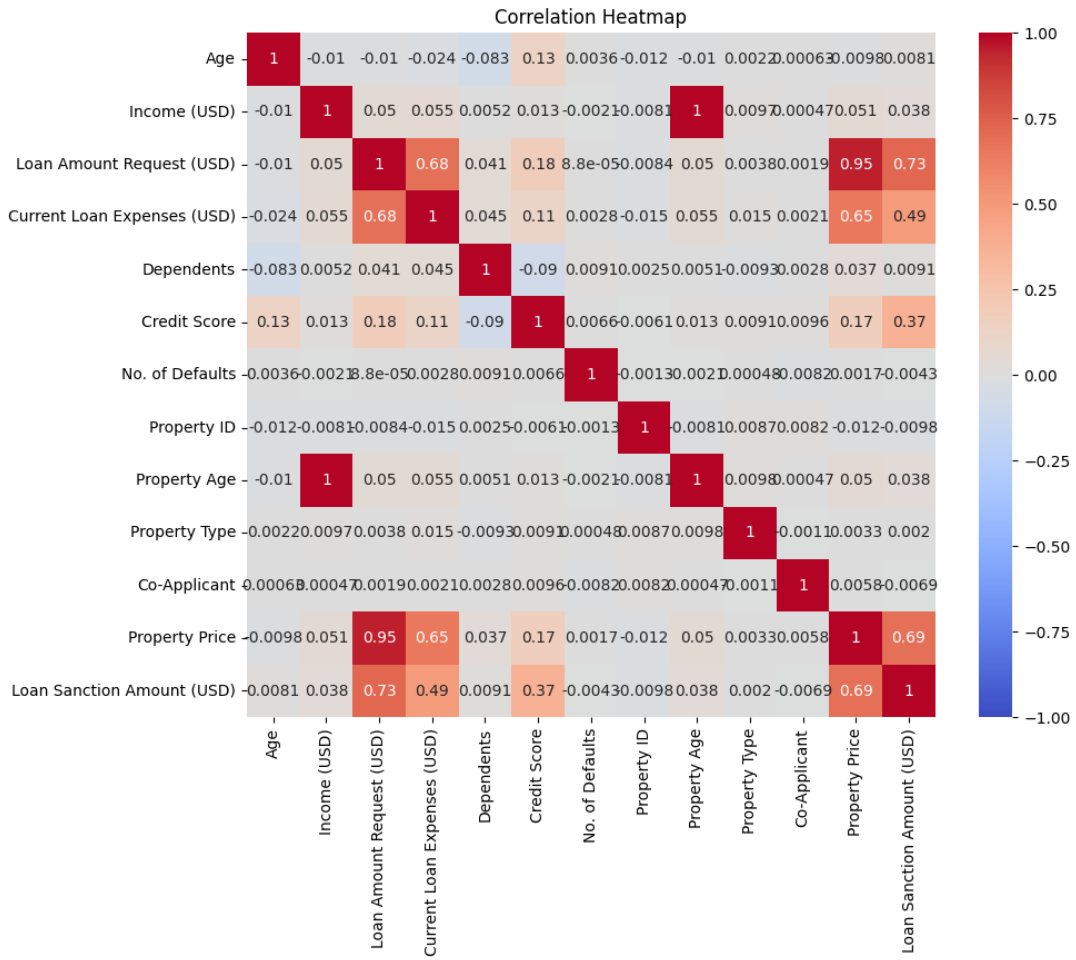

Distribution of Co-Applicant
Count
Co-Applicant

Distribution of Property Price
Count
Property Price
1e6


Distribution of Loan Sanction Amount (USD)
Count
Loan Sanction Amount (USD)

[6]:
```
if "Loan_Status" in df.columns:
    plt.figure(figsize=(12, 8))

    melted = df.melt(
        id_vars="Loan_Status",
        value_vars=numeric_cols,
        var_name="Feature",
        value_name="Value"
    )

    sns.boxplot(data=melted, x="Feature", y="Value", hue="Loan_Status")
    plt.title("Feature distributions by Loan Status")
    plt.xticks(rotation=45)
    plt.show()
```

[7]:
```
plt.figure(figsize=(10, 8))

corr = df[numeric_cols].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Correlation Heatmap")
plt.show()
```



#DIABETES DATASET

[8]:
```
df = pd.read_csv("diabetes.csv")
df.head()
```

[8]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

[9]:
```python
print("Shape:", df.shape)

print("\nInfo:")
df.info()

print("\nMissing values per column:")
print(df.isnull().sum())
```

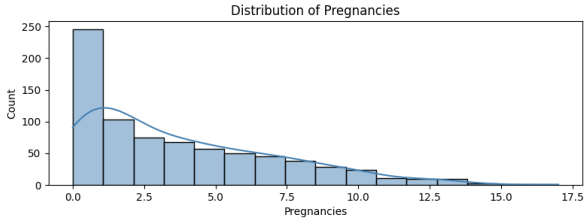
```
Shape: (768, 9)

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB

Missing values per column:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

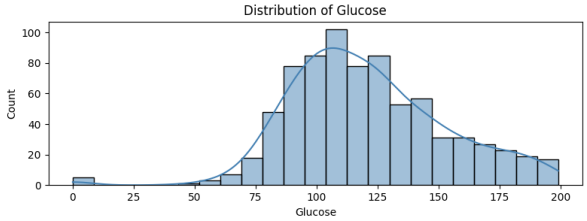
[10]:
```python
df.describe(include="all")
```

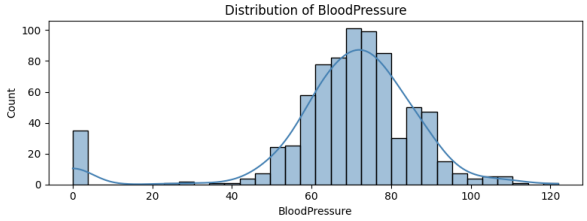
[10]:
```
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  120.894531      69.105469      20.536458   79.799479
```

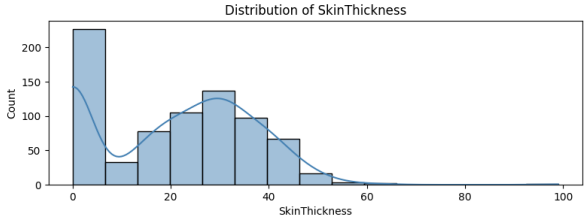
```
std        3.369578   31.972618      19.355807      15.952218  115.244002
min        0.000000    0.000000       0.000000       0.000000    0.000000
25%        1.000000   99.000000      62.000000       0.000000    0.000000
50%        3.000000  117.000000      72.000000      23.000000   30.500000
75%        6.000000  140.250000      80.000000      32.000000  127.250000
max       17.000000  199.000000     122.000000      99.000000  846.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    31.992578                  0.471876   33.240885    0.348958
std      7.884160                  0.331329   11.760232    0.476951
min      0.000000                  0.078000   21.000000    0.000000
25%     27.300000                  0.243750   24.000000    0.000000
50%     32.000000                  0.372500   29.000000    0.000000
75%     36.600000                  0.626250   41.000000    1.000000
max     67.100000                  2.420000   81.000000    1.000000
```

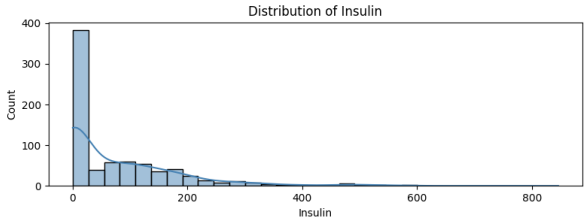
[11]:
```
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns

fig, axes = plt.subplots(len(numeric_cols), 1, figsize=(8,␣
 ↪3*len(numeric_cols)))

for i, col in enumerate(numeric_cols):
    sns.histplot(df[col], kde=True, ax=axes[i], color="steelblue")
    axes[i].set_title(f"Distribution of {col}")

plt.tight_layout()
plt.show()
```

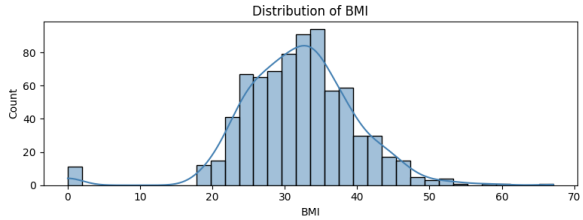

Distribution of Pregnancies
Count
Pregnancies

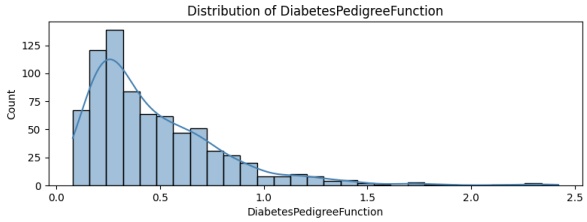

Distribution of Glucose
Count
Glucose

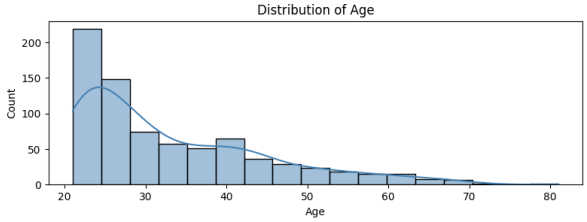

Distribution of BloodPressure
Count
BloodPressure

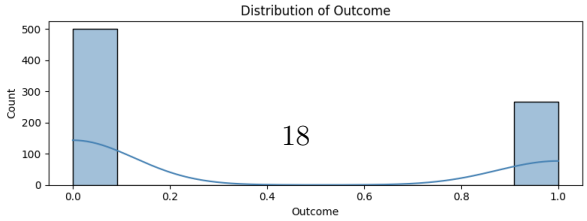

Distribution of SkinThickness
Count
SkinThickness


Distribution of Insulin
Count
Insulin

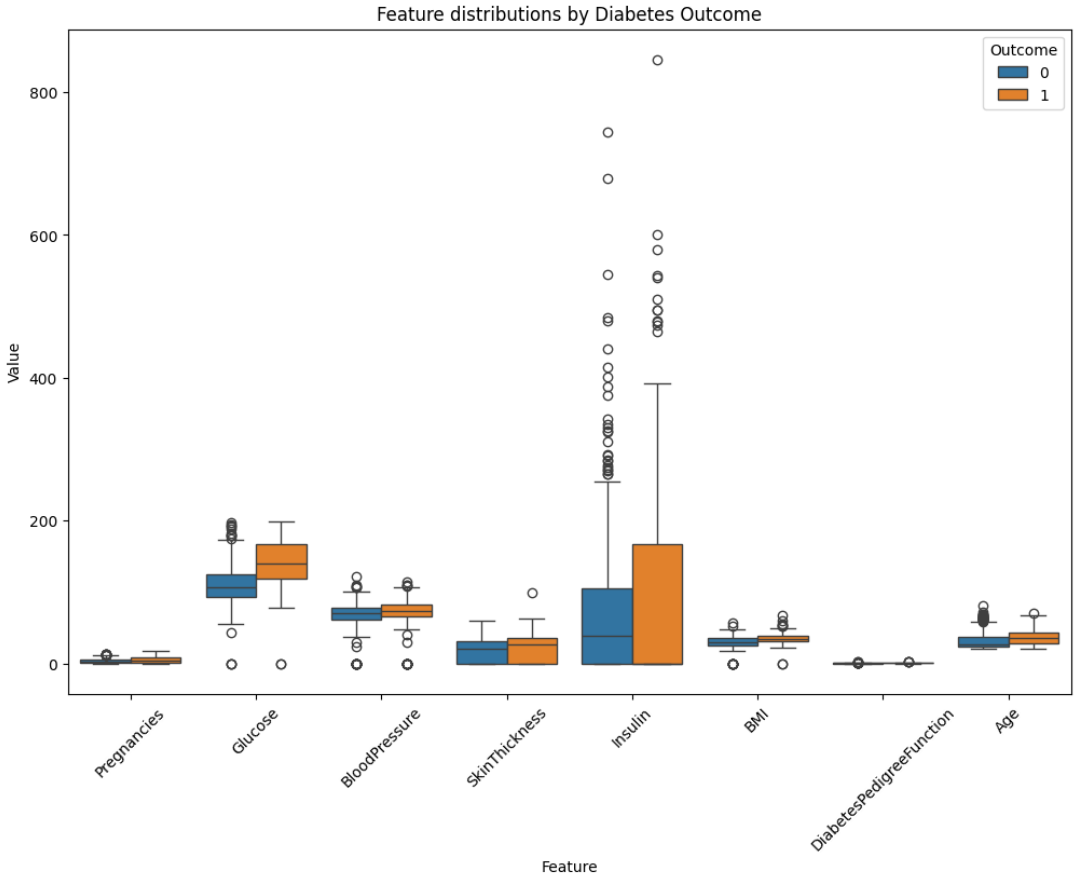

Distribution of BMI
Count
BMI


Distribution of DiabetesPedigreeFunction
Count
DiabetesPedigreeFunction


Distribution of Age
Count
Age

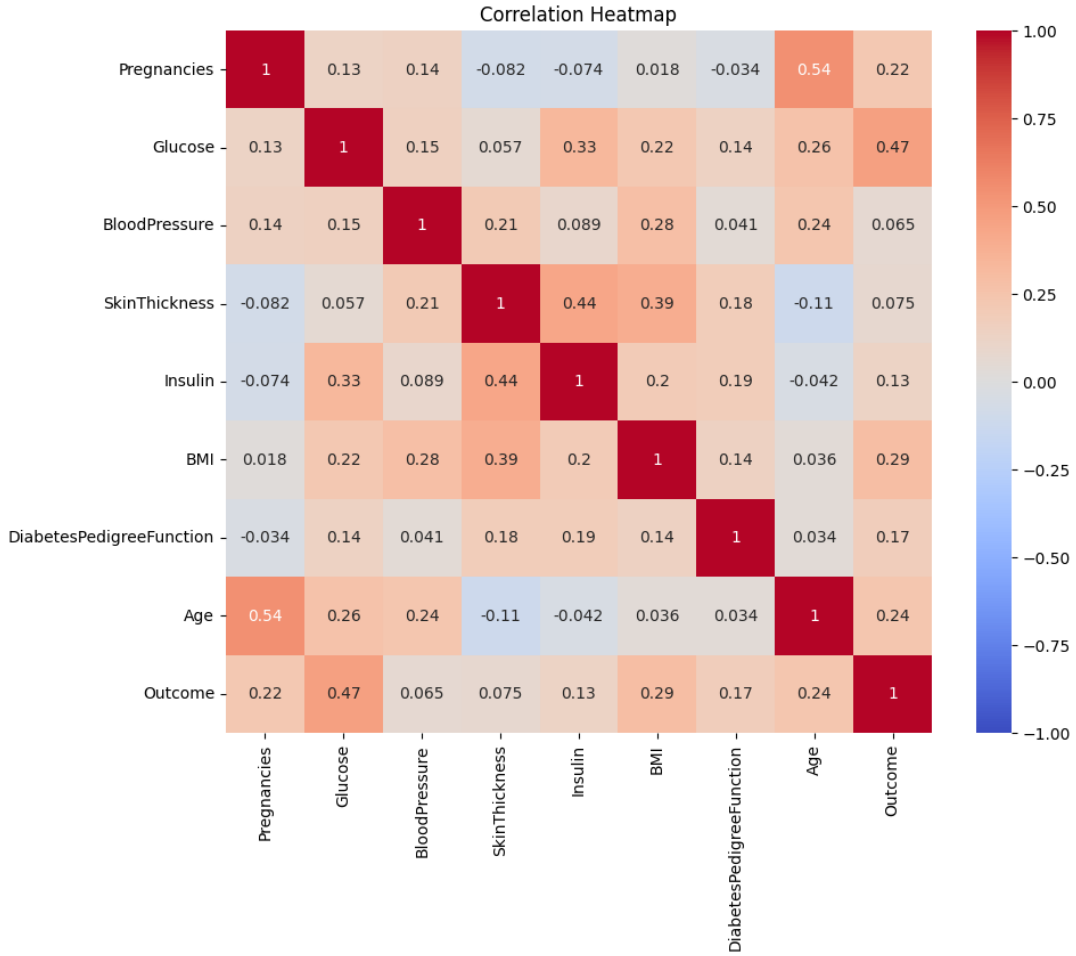

Distribution of Outcome
Count
Outcome

[12]:
```
if "Outcome" in df.columns:
    plt.figure(figsize=(12, 8))

    melted = df.melt(
        id_vars="Outcome",
        value_vars=numeric_cols,
        var_name="Feature",
        value_name="Value"
    )

    sns.boxplot(data=melted, x="Feature", y="Value", hue="Outcome")
    plt.title("Feature distributions by Diabetes Outcome")
    plt.xticks(rotation=45)
    plt.show()
```



[13]:
```
plt.figure(figsize=(10, 8))
```

```
corr = df[numeric_cols].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", vmin=-1, vmax=1)

plt.title("Correlation Heatmap")
plt.show()
```



[14]:
```
categorical_cols = df.select_dtypes(include=["object"]).columns

for col in categorical_cols:
    plt.figure(figsize=(6, 4))
    sns.countplot(data=df, x=col, hue="Outcome" if "Outcome" in df.columns␣
 ↪else None)
    plt.title(f"Counts of {col}")
    plt.xticks(rotation=45)
    plt.show()
```

#EMAIL CLASSIFICATION

[15]:
```
df = pd.read_csv("email.csv")
df.head()
```

[15]:
```
  Category                                            Message
0      ham  Go until jurong point, crazy.. Available only ...
1      ham                      Ok lar... Joking wif u oni...
2     spam  Free entry in 2 a wkly comp to win FA Cup fina...
3      ham  U dun say so early hor... U c already then say...
4      ham  Nah I don't think he goes to usf, he lives aro...
```

[16]:
```
print("Shape:", df.shape)
df.info()

print("\nMissing values:")
print(df.isnull().sum())
```

```
Shape: (5572, 2)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5572 entries, 0 to 5571
Data columns (total 2 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Category  5572 non-null   object
 1   Message   5572 non-null   object
dtypes: object(2)
memory usage: 87.2+ KB

Missing values:
Category    0
Message     0
dtype: int64
```

[17]:
```
df.describe(include="all")
```

[17]:
```
        Category                 Message
count       5572                    5572
unique         2                    5157
top          ham  Sorry, I'll call later
freq        4825                      30
```

[18]:
```
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns

if len(numeric_cols) > 0:
    rows = math.ceil(len(numeric_cols)/2)
    fig, axes = plt.subplots(rows, 2, figsize=(12, 4*rows))
    axes = axes.ravel()
```

```
    for i, col in enumerate(numeric_cols):
        sns.histplot(df[col], kde=True, ax=axes[i], color="steelblue")
        axes[i].set_title(f"Distribution of {col}")

    for j in range(i+1, len(axes)):
        fig.delaxes(axes[j])

    plt.tight_layout()
    plt.show()
```

[19]:
```
if len(numeric_cols) > 0:
    plt.figure(figsize=(10, 8))
    corr = df[numeric_cols].corr()
    sns.heatmap(corr, annot=True, cmap="coolwarm", vmin=-1, vmax=1)
    plt.title("Correlation Heatmap")
    plt.show()
```

#HANDWRITING RECOGNITION

[22]:
```
df = pd.read_csv("english.csv")
df.head()
```

[22]:

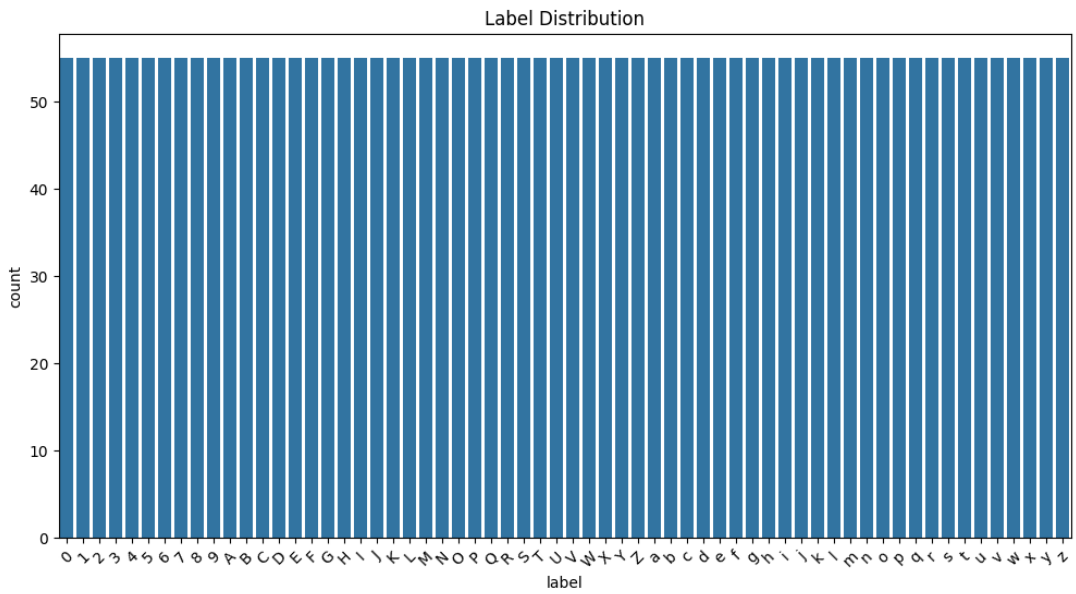
	image	label
0	Img/img001-001.png	0
1	Img/img001-002.png	0
2	Img/img001-003.png	0
3	Img/img001-004.png	0
4	Img/img001-005.png	0

[23]:
```
print("Shape:", df.shape)
df.info()

print("\nMissing values:")
print(df.isnull().sum())
```

```
Shape: (3410, 2)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3410 entries, 0 to 3409
Data columns (total 2 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   image   3410 non-null   object
 1   label   3410 non-null   object
dtypes: object(2)
memory usage: 53.4+ KB
```

```
Missing values:
image    0
label    0
dtype: int64
```

[24]:
```
plt.figure(figsize=(12, 6))
sns.countplot(data=df, x="label", order=df["label"].value_counts().index)
plt.title("Label Distribution")
plt.xticks(rotation=45)
plt.show()
```



[25]:
```
def show_samples(df, labels, samples_per_label=5):
    fig, axes = plt.subplots(
        len(labels), samples_per_label,
        figsize=(samples_per_label*2, len(labels)*2)
    )

    for i, label in enumerate(labels):
        subset = df[df["label"] == label].sample(samples_per_label,␣
↪random_state=42)
        for j, img_path in enumerate(subset["image"].values):
            path = "handwritten/" + img_path
            if os.path.exists(path):
                img = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
                axes[i, j].imshow(img, cmap="gray")
                axes[i, j].axis("off")
```

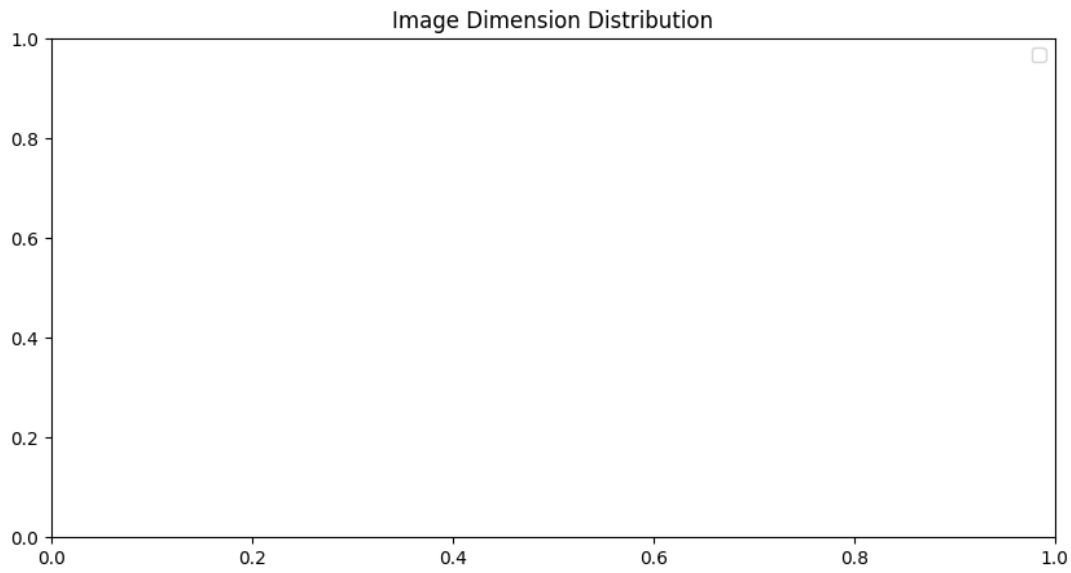
```
                axes[i, j].set_title(str(label))
            else:
                axes[i, j].text(0.5, 0.5, "Missing", ha="center", va="center")
                axes[i, j].axis("off")

    plt.tight_layout()
    plt.show()
```

[26]:
```
unique_labels = df["label"].unique()
show_samples(df, unique_labels[:5], samples_per_label=5)
```

[29]:
```
heights, widths = [], []

for img_path in df["image"].sample(min(200, len(df)), random_state=42):
    path = "handwritten/" + img_path
```

```
    if os.path.exists(path):
        img = cv2.imread(path)
        if img is not None:
            h, w = img.shape[:2]
            heights.append(h)
            widths.append(w)

plt.figure(figsize=(10, 5))

ax = sns.histplot(heights, kde=True, color="steelblue", label="Heights")
sns.histplot(widths, kde=True, color="orange", label="Widths", ax=ax)

ax.legend()
ax.set_title("Image Dimension Distribution")

plt.show()
```

```
C:\Users\Sabarithan P\AppData\Local\Temp\ipykernel_1392\2171688439.py:17:
UserWarning: No artists with labels found to put in legend.  Note that artists
whose label start with an underscore are ignored when legend() is called with no
argument.
  ax.legend()
```



[ ]:

```
```